Language-Guided Concept Bottlenecks for Speech Emotion Recognition

Shubha Sanket Samantaray & Kshitij Ambilduke
Multimodal Explainable AI Project
ENS Paris-Saclay

Abstract

In this work, we study *Language in a Bottle: Language Model Guided Concept Bottlenecks for Interpretable Image Classification* [1], which proposes an interpretable-by-design classifier whose predictions are mediated by natural-language concepts generated by a large language model. Although concept-based reasoning is attractive for explainability, standard bottleneck models rely on manually specified concepts and often underperform black-box baselines. LaBo shows that language-generated concepts, grounded by a pretrained multimodal encoder and used by a lightweight classifier, can remain competitive with end-to-end models.

We adapt this framework from image classification to speech emotion recognition on IEMOCAP [2] by replacing CLIP with CLAP [3], an audio-text model that provides a shared representation space for speech signals and textual descriptions. This yields an interpretable bottleneck of acoustic-emotional concepts such as *high vocal tension*, *flat intonation*, and *slow low-energy speech*. The report first explains the reference paper and then studies its adaptation to speech emotion recognition.

1 Relevant background

1.1 Motivation

The reference paper addresses a standard trade-off in explainable AI, highly accurate models are often opaque, while interpretable models are commonly viewed as too restrictive to remain competitive. Much of the explainable AI literature therefore focuses on *post-hoc* explanations, which describe a prediction after it has been made. Such explanations may be useful, but they need not faithfully reflect the model's actual reasoning [4]. The paper instead adopts *interpretability by design*, where the model itself is constrained to remain understandable.

The framework considered is that of *Concept Bottleneck Models* (CBMs) [5]. A CBM predicts a label through an intermediate set of semantically meaningful concepts rather than directly from latent features. This makes the decision process inspectable and, in principle, editable by a domain expert. However, classical CBMs have two major limitations; they require a manually specified concept vocabulary, and the resulting concept set is often incomplete, leading to weaker performance than unconstrained end-to-end models.

The central claim of *Language in a Bottle* is that both limitations can be addressed simultaneously. Instead of hand-designing concepts, the paper uses a large language model to generate them automatically; instead of learning concept detectors from scratch, it uses a pretrained multimodal alignment model to ground them in the input. The result is an automatically constructed concept bottleneck that remains interpretable while approaching black-box performance.

1.2 Methodology

The method, called LaBo, combines three ingredients, a language model, a multimodal encoder, and a linear concept-based classifier. For each class, the language model generates a large pool of candidate textual descriptions. A subset of these concepts is then selected by optimizing a criterion that favors both discriminative power and diversity. A pretrained vision-language model is used to score the presence of each selected concept in an image, and a linear classifier predicts the class label from the resulting concept activations.

The key novelty is that language is not only used for prompting or zero-shot labeling but also to *construct the explanatory layer itself.* The bottleneck therefore consists of natural-language units such as *brown head with white stripes* or *long thin tail*, rather than a small hand-crafted attribute list.

1.3 Formal Formulation

Let $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{n}$ be a labeled image classification dataset, where x_i is an image and $y_i \in \mathcal{Y}$ is a class label, with $|\mathcal{Y}| = N$. The paper assumes a pretrained vision-language model such as CLIP [6], with image encoder $\mathcal{I}$ and text encoder $\mathcal{T}$ mapping both modalities into a shared space $\mathbb{R}^d$.

Bottleneck construction. For each class $y \in \mathcal{Y}$, LaBo generates a class-specific candidate set S_y and selects exactly k concepts $C_y \subseteq S_y$. The full bottleneck is

$$C = \bigcup_{y \in \mathcal{Y}} C_y, \qquad |C| = N_C = N \times k,$$

so every class contributes equally to the final concept space.

Concept scoring. Each concept $c \in C$ is embedded by the text encoder, producing the concept matrix

$$E_C \in \mathbb{R}^{N_C \times d}, \quad (E_C)_j = \mathcal{T}(c_j)^\top.$$

For an image embedding $x = \mathcal{I}(i) \in \mathbb{R}^d$, the concept score vector is

$$g(x, C) = x E_C^\top \in \mathbb{R}^{N_C}.$$

Each entry of $g(x, C)$ is the dot-product alignment score between the image and one textual concept.

Label prediction. The final label is predicted from these concept scores through

$$\hat{y} = \arg\max\big(g(x, C)\, \sigma(W)^\top\big),$$

where $W \in \mathbb{R}^{N \times N_C}$ is a learnable class-concept weight matrix and $\sigma(\cdot)$ is a softmax normalization across classes for each concept:

$$\sigma(W)_{y,c} = \frac{e^{W_{y,c}}}{\sum_{y' \in \mathcal{Y}} e^{W_{y',c}}}.$$

This makes the class-concept association easier to interpret. Since there is no residual path from image features to labels, the model must predict through the concept layer.

1.4 Concept Generation

The first stage of LaBo generates candidate concepts with GPT-3, specifically `text-davinci-002`. For each class, the model is queried with five prompt templates, the main one being *describe what the [class] looks like.* For fine-grained datasets, an optional super-class qualifier is added to reduce ambiguity, and GPT-3 is queried until 500 raw sentences are collected per class.

These raw sentences are then decomposed into shorter concept phrases by a T5-large model [9] fine-tuned for concept extraction. The fine-tuning data consists of 500 annotated sentence-concept pairs, obtained by sampling 100 examples from each of five datasets. For example, a sentence such as *the axolotl's limbs are delicate, and the tail is long and thin* is split into concepts such as *limbs are delicate* and *tail is long and thin.* The paper then removes class-name leakage in two stages: full class names are replaced by a super-class name when possible, and concepts containing all tokens of a multi-word class name are discarded. This step helps convert free-form language-model output into short, label-free bottleneck units.

1.5 Submodular Concept Selection

Since each class yields a large candidate pool S_y, the paper selects a subset $C_y \subseteq S_y$ of size k by maximizing

$$\mathcal{F}(C_y) = \alpha \sum_{c \in C_y} D(c) + \beta \sum_{s \in S_y} \max_{c \in C_y} \phi(s, c),$$

where $D(c)$ is a *discriminability* score and $\phi(s, c)$ is a similarity-based *coverage* term. Because $\mathcal{F}$ is a non-negative linear combination of monotone submodular functions, it is itself monotone submodular, so greedy optimization admits a constant-factor approximation guarantee [12].

Discriminability. Let

$$\mathrm{Sim}(y, c) = \frac{1}{|\mathcal{X}_y|} \sum_{x \in \mathcal{X}_y} x \cdot \mathcal{T}(c)^\top$$

be the mean image-text alignment between class y and concept c, where $\mathcal{X}_y$ is the set of training embeddings for class y. The normalized class association is

$$\overline{\mathrm{Sim}}(y \mid c) = \frac{\mathrm{Sim}(y, c)}{\sum_{y' \in \mathcal{Y}} \mathrm{Sim}(y', c)},$$

and the discriminability score is

$$D(c) = \sum_{y' \in \mathcal{Y}} \overline{\mathrm{Sim}}(y' \mid c) \log \overline{\mathrm{Sim}}(y' \mid c).$$

This is the negative entropy of $\overline{\mathrm{Sim}}(\cdot \mid c)$, so maximizing $D(c)$ selects concepts whose alignment is concentrated on one or a few classes. Concepts that align similarly with all classes receive a low score.

Coverage. The second term encourages semantic diversity within the selected subset. The pairwise similarity between concepts is measured by cosine similarity between text embeddings:

$$\phi(c_1, c_2) = \cos(\mathcal{T}(c_1), \mathcal{T}(c_2)).$$

Maximizing the facility-location term prevents the bottleneck from collapsing onto several near-paraphrases of the same attribute. The hyperparameters α and β balance discriminability and diversity and are tuned on the development set.

1.6 Prediction and Language Prior

Once the bottleneck concepts are selected, CLIP is used as a frozen alignment model to score their presence in each image. These concept scores are then fed to the linear classifier. The weight matrix W is not initialized randomly; instead, the paper uses the binary language prior

$$W_{y,c} = \begin{cases} 1 & \text{if } c \in C_y, \\ 0 & \text{otherwise,} \end{cases}$$

so that concepts selected for a class begin with a stronger association to that class. This is especially useful in few-shot settings, where labeled data is insufficient to learn reliable class-concept weights from scratch.

Language therefore contributes twice, first by generating and selecting the bottleneck concepts, and second by biasing the optimization of the final classifier.

1.7 Empirical results

LABO is evaluated across numerous image classification datasets spanning common objects, fine-grained categories, actions, textures, skin lesions, and satellite imagery. The method is compared against several main baselines namely:

- **Linear Probe:** This serves as the primary black-box reference, utilizing a logistic regression classifier trained on frozen CLIP image embeddings.
- **PCBM (Post-hoc Concept Bottleneck Model):** PCBM addresses the same weaknesses as LaBo but uses a different strategy by incorporating ConceptNet concepts and adding a residual path from image features directly to the labels, which weakens the strict bottleneck interpretation. LaBo clearly outperforms standard PCBM on common object classification tasks and remains highly competitive with PCBM's stronger residual variant.
- **CompDL (Compositional Derivation Learning):** This baseline learns a linear layer over CLIP similarities between human-designed class descriptions and images. LaBo significantly outperforms CompDL on fine-grained categorization tasks across varying levels of supervision, and notably achieves

this without relying on manual concepts.

Overall, when compared to the primary Linear Probe baseline, LABO demonstrates strong improvements in low-data settings and shows better average performance across the datasets. However, in fully supervised scenarios, LABO marginally underperforms the Linear Probe. This slight drop in peak performance is expected, as LABO's bottleneck architecture imposes a stricter structural constraint than a standard black-box model.

2 Our contributions

Our project adapts LaBo from image classification to speech emotion recognition. The target task is utterance-level emotion classification on IEMOCAP [2]. The question is whether language-guided concept bottlenecks remain both interpretable and predictively useful when speech, a modality inherently carrying finer granularity of information, is in picture.

Emotion labels are ambiguous, speaker-dependent, and sensitive. A model that predicts *angry* or *sad* without being transparent about the the cues on which it relies is difficult to trust in any high-stakes deployment. In contrast, a concept bottleneck model can make its reasoning visible through interpretable textual descriptions such as *tense and raised voice*, *slow speech punctuated by pauses*, or *high energy with abrupt emphasis*.

2.1 Task and Dataset

We use IEMOCAP [2], a standard benchmark for speech emotion recognition. The corpus consists of ten actors performing scripted and improvised interactions across several recording sessions. Each session is segmented into utterances and annotated with emotion labels by multiple evaluators. The dataset provides train and test set hence we keep 10% of training set for validation. We notice a rather surprisingly skewed distribution of examples across classes, however, we do not re-sample the dataset in any of our experiments in order to test LaBo in constrained scenarios instead of ideal ones. The overall statistics of the datasets can be seen in Table 1. We report the accuracy on the test set as the main evaluation metric.

Class	**Count**
frustrated	2917
excited	1976
neutral	1726
angry	1269
sad	1250
happy	656
surprise	110
fear	107
other	26
disgust	2

Table 1: Distribution of utterances across classes in the IEMOCAP dataset

2.2 Replacing CLIP with CLAP

The key architectural substitution is the replacement of CLIP by CLAP [3]. In the original paper, CLIP provides a shared image-text embedding space in which concept scores are computed by image-text dot product similarity. For audio, the natural parallel is CLAP [3], a contrastive model trained on large-scale audio-text pairs that maps audio clips and textual descriptions into a common representation space via similar objectives to CLIP.

Formally, let $\mathcal{A}$ denote the CLAP audio encoder and $\mathcal{T}$ its text encoder. Both encoders are used with frozen weights throughout, no fine-tuning of the audio or text encoder is performed at any stage. This preserves the zero-shot grounding property of the original method and keeps the bottleneck strictly decoupled from the downstream classification loss.

For an utterance waveform u, we pre-extract its audio embedding once:

$$a = \mathcal{A}(u) \in \mathbb{R}^d.$$

For a concept c, the textual embedding is

$$t_c = \mathcal{T}(c) \in \mathbb{R}^d.$$

Given the selected bottleneck $C = \{c_1, \ldots, c_{N_C}\}$, the concept matrix is

$$E_C = \begin{bmatrix} t_{c_1}^\top \\ \vdots \\ t_{c_{N_C}}^\top \end{bmatrix} \in \mathbb{R}^{N_C \times d}.$$

The concept scores for an utterance are then

$$g(u, C) = aE_C^\top \in \mathbb{R}^{N_C},$$

exactly mirroring the image-text scoring of the original formulation. Each coordinate of this vector measures the degree to which the utterance matches one textual acoustic-emotional concept. The final prediction is made from these scores through a linear classifier with softmax-normalized class-concept weights, preserving the strict bottleneck structure of LaBo. Pre-extracting and caching all audio embeddings before training makes the computational overhead of this approach negligible, training the linear layer on top of frozen CLAP embeddings takes much lesser time per epoch as compared to extracting embeddings on the fly.

2.3 Candidate Concept Generation

Adapting LaBo to speech requires a fundamental redefinition of what is considered as a concept. In image classification, concepts refer to visible attributes such as color, shape, or texture. In speech emotion recognition, the relevant explanatory units are acoustic and paralinguistic. We therefore construct a concept vocabulary centered on *Prosody*, *Energy*, *Voice quality* and *Expressive style*. We are not simply applying the same architecture to another encoder, we are reformulating the concept space around interpretable descriptors that are both meaningful for emotional speech and sufficiently grounded in CLAP's representation space.

Following the original paper, we use a language model to generate a large pool of class-specific candidate concepts. We use Claude Sonnet 4.6 as the concept generator. Unlike the image setting, where concepts describe static visual attributes, our concepts must describe dynamic, temporal, and paralinguistic properties of speech, which requires a more capable model to produce reliably non-trivial descriptions.

The prompts are adapted to the audio-emotion setting. We use three primary templates:

- Describe the vocal and acoustic characteristics of a person who is `[emotion]`.
- What prosodic and paralinguistic cues indicate that a speaker is feeling `[emotion]`?
- Describe the speaking rate, pitch, voice quality, and energy of someone who is `[emotion]`.

Caption Generation and Reranking. To construct our final concept pool, we employed a two-step prompting strategy. First, we instructed a Large Language Model (Claude Sonnet 4.6) to generate an initial diverse set of 15 candidate captions for each emotion class. Second, we prompted the same LLM to evaluate and rerank these generated captions based on their acoustic accuracy, descriptiveness, and distinctiveness. Leveraging the same LLM for both generation and subsequent reranking has been shown to significantly improve final output quality by separating the generative and evaluative steps, the model can critically assess its own outputs [10, 11]. Following this reranking process, we filtered the candidates to select only the top 8 captions per class. The final selected prompts used for downstream processing can be found in the appendix.

2.4 Final Concept Bottleneck Classifier

After bottleneck construction, each utterance is represented exclusively by its concept activation vector $g(u, C) \in \mathbb{R}^{N_C}$. A linear classifier is then trained on top of these concept scores, using the Adam optimizer with a sweep over learning rates and batch sizes on the validation fold.

The class-concept weight matrix is initialized with the language prior of the original paper: $W_{y,c} = 1$ if $c \in C_y$, else 0. This gives concepts generated for a given emotion a head start in the optimization, which is particularly valuable in few-shot conditions where the training signal is sparse.

We deliberately keep this final stage simple. The purpose of the project is not to maximize raw benchmark performance with a complex downstream architecture, but to test whether an emotion classifier can remain interpretable while relying exclusively on concept activations. The linear layer therefore has a clear semantic role, its weight $W_{y,c}$ measures the degree to which concept c supports the prediction of emotion y, and this is directly inspectable.

3 Results

We evaluate five approaches to speech emotion recognition on the IEMOCAP dataset: a zero-shot CLAP baseline, a k-nearest neighbour (k-NN) retrieval baseline, an end-to-end CNN trained on mel spectrograms (E2E), a frozen CLAP linear probe (CLAP Probe), and a language-bottleneck model (LaBo). We begin by describing each method, then report quantitative results, and finally provide a qualitative analysis of learned representations and failure modes.

3.1 Baselines

Zero-shot CLAP. The zero-shot baseline uses the joint audio-text embedding space learned by CLAP [3] without any task-specific training. At inference time, a set of text prompts is constructed for each emotion class and the audio embedding of an utterance is compared against these text embeddings via cosine similarity; the class whose prompt is most similar to the audio is predicted.

k-Nearest Neighbours (k-NN). The k-NN baseline uses frozen CLAP audio embeddings as a feature extractor without any fine-tuning. All training utterances are embedded and stored. At test time, the k most similar training embeddings (by cosine similarity) are retrieved and the majority class label is returned. This baseline measures how linearly separable the raw CLAP audio space is for emotion classification.

CLAP Probe. The CLAP Probe keeps the CLAP encoder frozen and trains a lightweight two-layer MLP head on top of the audio embeddings. This isolates the quality of the fixed CLAP representations while allowing a learned, non-linear decision boundary. It serves as an upper bound for methods that rely purely on frozen CLAP features.

LaBo. Language Bottleneck, as discussed before, includes an explicit concept layer between the audio encoder and the classifier. For each emotion class, a set of natural-language concept descriptions is embedded with the CLAP text encoder. An audio utterance is scored against every concept via cosine similarity, and a learned weighting matrix W aggregates these concept scores into class logits. The model is interpretable by design; the weight of each concept in W reveals which acoustic descriptions the classifier relies upon.

End-to-end CNN (E2E). As an additional point of comparison, we train a four-block CNN directly on log-mel spectrograms, with no pre-training. This baseline measures how much of CLAP's benefit comes from its pre-trained representations versus a simple supervised feature extractor.

3.2 Quantitative results

Zero-shot aggregation strategies. We evaluate three ways of constructing per-class text representations. The *single* strategy uses one manually written prompt per class (*"a person speaking with {emotion} emotion"*). The *mean* strategy embeds a bank of eight acoustically-grounded concept descriptions per class and averages their text embeddings before computing similarity. The *max* strategy selects the highest similarity score across all concepts in a class's bank. All three strategies perform as well as random guessing. This suggests that the CLAP text-audio alignment is not reliable enough for zero-shot emotion discrimination off-the-shelf. This is contrary to how well CLIP, an image-to-text alignment model, works in zero-shot settings.

Method	**Test Accuracy (%)**
Zero-shot CLAP (single prompt)	7.97
Zero-shot CLAP (mean aggregation)	9.06
Zero-shot CLAP (max aggregation)	8.37
k-NN (k=5)	35.46
LaBo	33.85
CLAP Probe	44.12
E2E CNN	**44.42**

Table 2: Test accuracy on IEMOCAP. Higher is better.

The zero-shot baselines perform at chance, indicating that CLAP's joint embedding space does not encode emotion in a way that is usable without supervision. The k-NN baseline, on the other hand, shows that the raw CLAP audio embeddings do carry some emotion-discriminative signal, even though that signal is not accessible via text-side matching. Interestingly, k-NN outperforms LaBo. This suggests that routing predictions through alignment scores degrades the signal rather than refining it, and this is consistent with CLAP not being a reliable alignment model for fine-grained emotion discrimination.

The two trained models on top of frozen CLAP features, namely CLAP Probe and E2E CNN, perform comparably and substantially outperform all other methods. The competitive performance of the E2E CNN, trained from scratch with no pre-training shows that CLAP's representations offer limited advantage over raw mel spectrograms for the task of emotion classification.

3.3 Hyperparameter Sensitivity

We performed a grid search over learning rate and number of concepts per class for LaBo, using 15 training epochs per configuration. Table 3 reports validation accuracy across the full grid.

Learning Rate	**Concepts per Class**	**Val. Accuracy (%)**
1×10^{-3}	5	33.85
1×10^{-3}	8	33.85
1×10^{-3}	3	33.74
3×10^{-4}	3	32.19
3×10^{-4}	5	31.97
3×10^{-4}	8	30.97
1×10^{-4}	3	27.77
1×10^{-4}	5	26.99
1×10^{-4}	8	25.22

Table 3: LaBo hyperparameter sweep. Reporting validation accuracy across learning rate and number of concepts per class.

The sweep reveals two consistent trends. First, a higher learning rate of 1×10^{-3} yields the best validation accuracy across all concept bank sizes, with lower learning rates leading to monotonically worse performance. Second, the number of concepts per class has negligible impact at the best learning rate (33.85% for both 5 and 8 concepts), but at lower learning rates, smaller concept banks tend to outperform larger ones. This is consistent with a model that relies on a narrow, well-aligned subset of concepts, adding more concepts introduces noise rather than discriminative signal when the weighting matrix has not converged. The best configuration ($lr = 10^{-3}$, $k = 8$) was used for the test-set evaluation in Table 3.2.

3.4 Qualitative analysis

Concept weight heatmap. Figure 1 visualises the learned LaBo weight matrix W after applying the column-wise softmax used during the forward pass. Each row corresponds to an emotion class and each column to a concept. For several classes like *angry*, *excited*, *frustrated*, *sad*, and *neutral* the highest weights are concentrated on appropriate concepts. For example, the top-weighted concepts

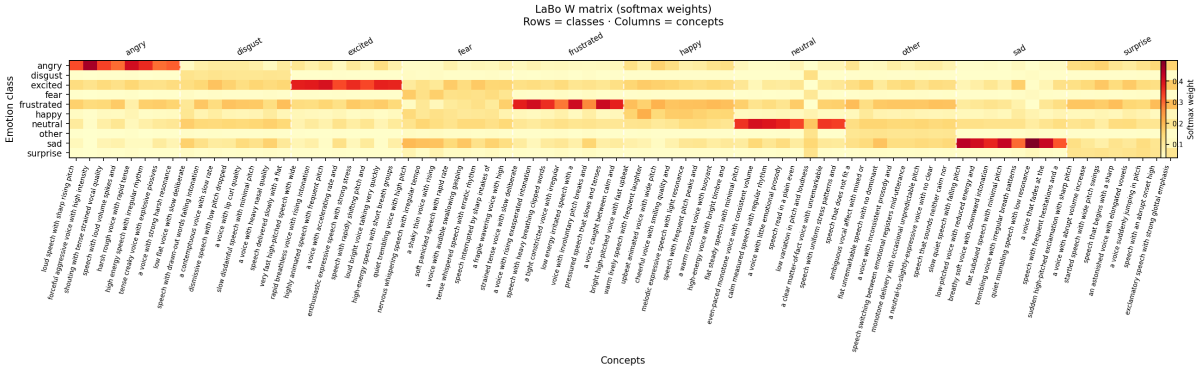

Figure 1: Learned LaBo weight matrix W after column-wise softmax. Rows correspond to emotion classes and columns correspond to concepts. Concentrated bright cells for *angry*, *excited*, *frustrated*, *sad*, and *neutral* indicate that the model identifies acoustically appropriate concepts for these classes.

for *angry* consistently reference harsh, high-intensity speech, while those for *sad* capture low-energy, falling-intonation descriptions. Contrastively, classes such as *fear*, *disgust*, and *surprise* show more random weight distributions, suggesting that CLAP's text encoder does not produce well-separated embeddings for these more ambiguous acoustic categories. This inconsistency across classes partly explains LaBo's overall accuracy.

Concept alignment vs. top-8 retrieved concepts. Figure 2 shows, for each class, the mean cosine similarity between the class's audio embeddings and each concept in its designated concept block. Ideally, if CLAP's audio-text alignment were well-calibrated for emotion, the top-scoring concepts within each class block should coincide with the globally top-scoring concepts across all classes. In practice, we observe a discrepancy. The concept descriptions that rank highest within a class's own block are often very different from those that achieve the highest alignment scores across the entire concept bank. This indicates that CLAP's audio-text similarity is dominated by low-level acoustic surface features rather than emotion-specific semantics, and that the model cannot reliably distinguish which concept is most appropriate for a given utterance.

Confusion matrices. Figure 4 shows recall confusion matrices for LaBo, k-NN, and CLAP Probe. The impact of class imbalance is immediately apparent. LaBo collapses almost entirely to predicting the single most frequent class, producing near-zero recall on all minority classes. This failure mode is consistent with the alignment-score bottleneck providing an insufficiently discriminative signal once the weighting matrix is drawn towards the dominant class by the cross-entropy gradient. In contrast, both k-NN and CLAP Probe distribute predictions more evenly across classes. CLAP Probe achieves the most balanced confusion matrix, with meaningful recall on all four major emotion categories, though confusions between acoustically similar classes (e.g., *angry* and *excited*, or *sad* and *neutral*) remain the primary source of error.

4 Discussion

LaBo's design rests on two assumptions that hold well in the image domain but become fragile in speech. First, the method depends on a language model generating concepts that are both acoustically specific and reliably groundable by the multimodal encoder. In practice, emotional speech concepts are relatively abstract. They lie between low-level signal properties, such as pitch, energy, and rhythm, and higher-level psychological states, which makes them considerably harder to describe in text than visual attributes such as reddish-brown fur or a long, thin tail. Second, the scoring mechanism assumes the encoder's embedding space is well-calibrated for the relevant signal. This alignment between language knowledge and multimodal representations is useful only when that calibration holds, and our experiments show it does not hold reliably for CLAP in the emotion domain.

This adaptation is therefore more than a simple change of modality. In image classification, relevant

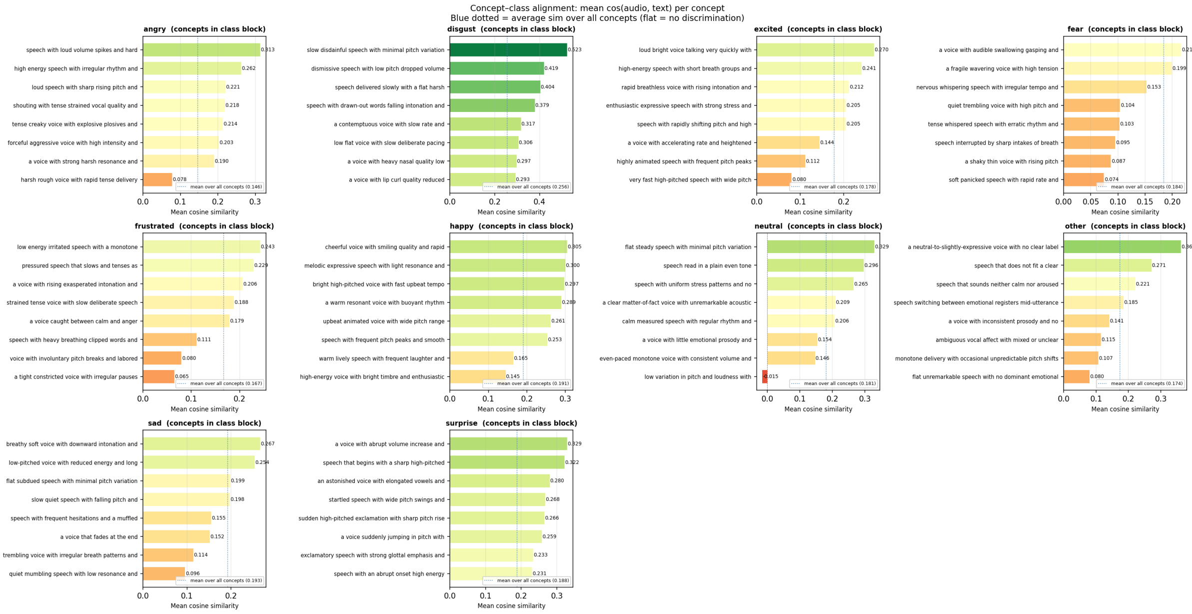

Figure 2: **Concept alignment.** For each emotion class, we plot the mean cosine similarity between the class's audio embeddings and every concept in its designated concept block, sorted in descending order. The blue dotted line marks the mean similarity averaged over all concepts in the block.

cues are often spatially localized and relatively stable. On the other hand, in emotional speech, they are temporally spread across the utterance and may be mixed with speaker identity, recording conditions, and spoken content. IEMOCAP also introduces strong speaker-level confounding. Since the dataset contains only a small number of actors, emotion labels can become closely tied to individual vocal styles. This makes a concept bottleneck useful as a diagnostic tool, because the selected concepts and their weights can be inspected directly. However, this interpretation is only meaningful if the alignment scores are reliable enough to preserve discriminative information through the bottleneck.

5 Conclusion

Our results point to an important difference between the two modalities. Visual concepts can often be described in text more directly than acoustic-emotional concepts, and CLAP does not appear to provide the same level of zero-shot alignment as CLIP. The zero-shot baselines perform at chance; k-NN outperforms LaBo, indicating that routing predictions through concept alignment scores degrades the available signal rather than refining it. At the same time, the learned weight matrix W suggests that the learning component is behaving reasonably. For classes where CLAP alignment is at least partly informative, such as angry, sad, frustrated, excited, and neutral, the model identifies concept-class associations that are acoustically coherent. The deeper failure lies in the quality of the concept activations themselves. Class imbalance makes this problem even more pronounced. Unlike the CLAP Probe or k-NN, LaBo tends to collapse toward the dominant class, because the bottleneck removes much of the discriminative signal needed to distinguish minority emotions. Still, the interpretability goal of the framework remains well motivated. Inspecting the concepts that influence a prediction, and assessing whether they are prosodically meaningful, provides the kind of transparency needed in high-stakes emotion recognition. In the audio setting, however, this will likely require either a stronger audio-text alignment model or a concept vocabulary designed more directly around the structure of CLAP's embedding space.

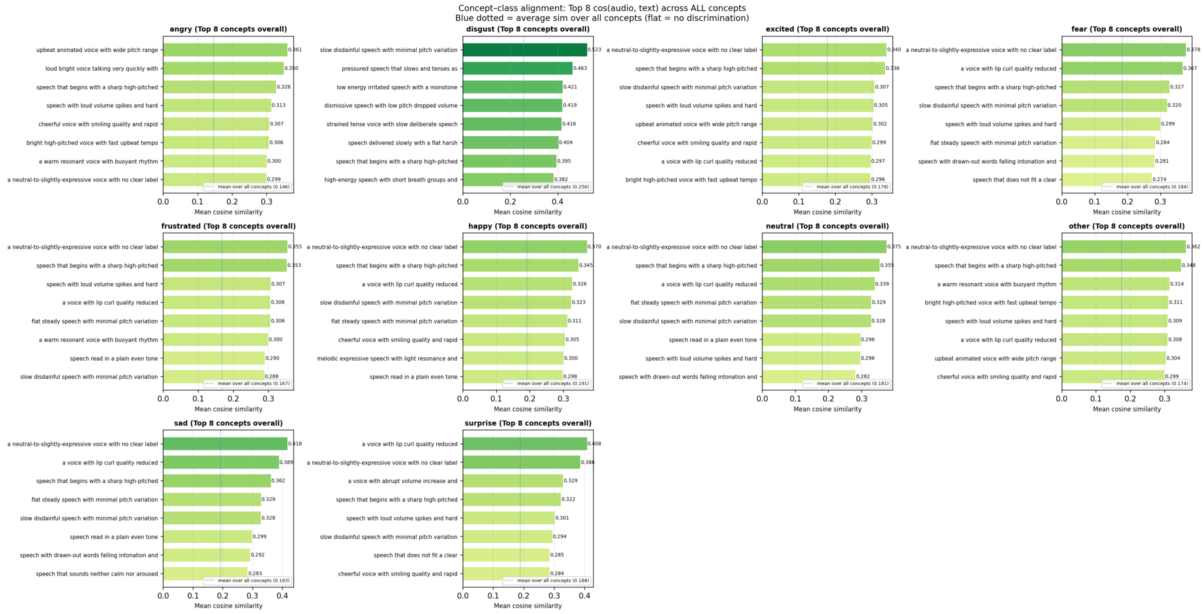

Figure 3: **Top-8 concept alignment.** For each class, the eight concepts with the highest mean cosine similarity to the class's audio embeddings are shown, drawn from the *full* concept bank across all classes.

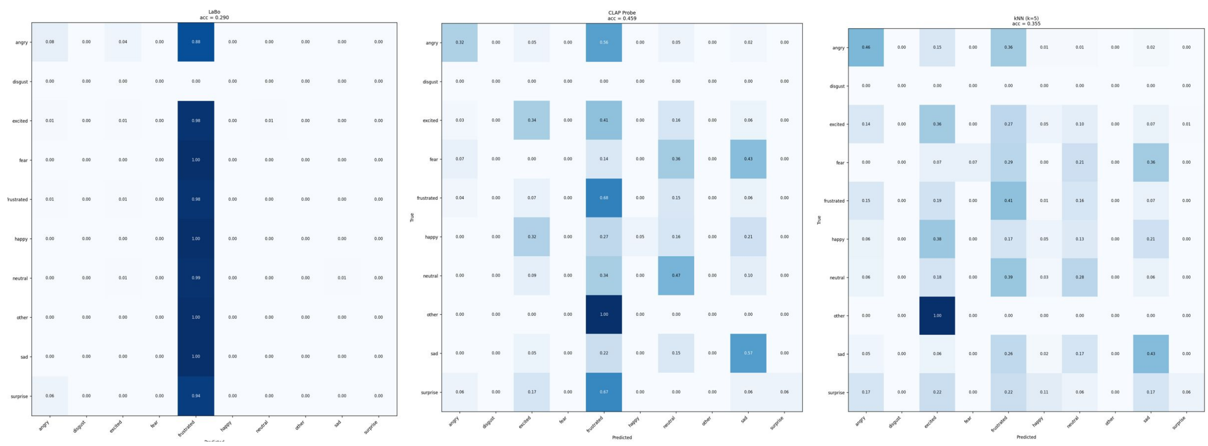

Figure 4: Confusion matrices for LaBo, k-NN (k=5), and CLAP Probe.

Appendix

A Concepts

```
"angry": [
  "loud speech with sharp rising pitch and clipped harsh consonants",
  "forceful aggressive voice with high intensity and fast rate",
  "shouting with tense strained vocal quality and abrupt pauses",
  "speech with loud volume spikes and hard glottal attacks",
  "harsh rough voice with rapid tense delivery and strong stress",
  "high energy speech with irregular rhythm and loud bursts",
  "tense creaky voice with explosive plosives and elevated pitch",
  "a voice with strong harsh resonance and confrontational tone"
],
"happy": [
  "bright high-pitched voice with fast upbeat tempo and rising intonation",
  "warm lively speech with frequent laughter and light breathy quality",
  "upbeat animated voice with wide pitch range and energetic delivery",
  "cheerful voice with smiling quality and rapid speech rate",
  "melodic expressive speech with light resonance and joyful inflections",
  "speech with frequent pitch peaks and smooth flowing rhythm",
  "a warm resonant voice with buoyant rhythm and positive energy",
  "high-energy voice with bright timbre and enthusiastic cadence"
],
"sad": [
  "slow quiet speech with falling pitch and monotone delivery",
  "low-pitched voice with reduced energy and long pauses between words",
  "breathy soft voice with downward intonation and slow tempo",
  "flat subdued speech with minimal pitch variation and weak intensity",
  "trembling voice with irregular breath patterns and slow rate",
  "quiet mumbling speech with low resonance and dragging rhythm",
  "a voice that fades at the end of phrases with low energy",
  "speech with frequent hesitations and a muffled mournful quality"
],
"neutral": [
  "flat steady speech with minimal pitch variation and moderate tempo",
  "even-paced monotone voice with consistent volume and no strong inflection",
  "calm measured speech with regular rhythm and neutral resonance",
  "a voice with little emotional prosody and steady moderate pitch",
  "speech read in a plain even tone without emphasis or energy peaks",
  "low variation in pitch and loudness with a consistent speaking rate",
  "a clear matter-of-fact voice with unremarkable acoustic properties",
  "speech with uniform stress patterns and no prominent prosodic features"
],
"frustrated": [
  "strained tense voice with slow deliberate speech and audible sighing",
  "a voice with rising exasperated intonation and drawn-out words",
  "speech with heavy breathing clipped words and forced controlled tempo",
  "a tight constricted voice with irregular pauses and rising pitch",
  "low energy irritated speech with a monotone edge and trailing volume",
  "voice with involuntary pitch breaks and labored breath patterns",
  "pressured speech that slows and tenses as if suppressing emotion",
  "a voice caught between calm and anger with strained resonance"
],
"excited": [
  "very fast high-pitched speech with wide pitch range and loud volume",
  "rapid breathless voice with rising intonation and energetic bursts",
  "highly animated speech with frequent pitch peaks and fast tempo",
  "a voice with accelerating rate and heightened breathiness",
  "enthusiastic expressive speech with strong stress and lively rhythm",
  "speech with rapidly shifting pitch and high acoustic energy",
  "loud bright voice talking very quickly with great enthusiasm",
  "high-energy speech with short breath groups and emphatic stress"
],
"fear": [
  "quiet trembling voice with high pitch and rapid shallow breathing",
  "nervous whispering speech with irregular tempo and breathy quality",
  "a shaky thin voice with rising pitch and frequent hesitations",
  "soft panicked speech with rapid rate and unstable vocal quality",
  "a voice with audible swallowing gasping and strained high pitch",
  "tense whispered speech with erratic rhythm and low volume",
  "speech interrupted by sharp intakes of breath and trembling pitch",
  "a fragile wavering voice with high tension and constricted resonance"
],
"surprise": [
  "sudden high-pitched exclamation with sharp pitch rise and loud onset",
  "a voice with abrupt volume increase and rapid pitch jump",
  "startled speech with wide pitch swings and gasping intake of breath",
  "speech that begins with a sharp high-pitched burst and then slows",
  "an astonished voice with elongated vowels and rising intonation",
  "a voice suddenly jumping in pitch with a loud breathy exclamation",
  "speech with an abrupt onset high energy burst and falling tail",
  "exclamatory speech with strong glottal emphasis and wide dynamic range"
],
"disgust": [
  "low flat voice with slow deliberate pacing and nasal resonance",
  "speech with drawn-out words falling intonation and lip tension",
  "a contemptuous voice with slow rate and harsh vocal texture",
  "dismissive speech with low pitch dropped volume and clipped endings",
  "a voice with lip curl quality reduced resonance and flat affect",
  "slow disdainful speech with minimal pitch variation and cold tone",
  "a voice with heavy nasal quality low energy and flat cadence",
  "speech delivered slowly with a flat harsh edge and low intensity"
],
"other": [
  "speech that does not fit a clear emotional category",
  "ambiguous vocal affect with mixed or unclear emotional tone",
  "flat unremarkable speech with no dominant emotional quality",
  "a voice with inconsistent prosody and no single prevailing emotion",
  "speech switching between emotional registers mid-utterance",
  "monotone delivery with occasional unpredictable pitch shifts",
  "a neutral-to-slightly-expressive voice with no clear label",
  "speech that sounds neither calm nor aroused with weak affect"
]
```

References

[1] Yue Yang, Artemis Panagopoulou, Shenghao Zhou, Daniel Jin, Chris Callison-Burch, and Mark Yatskar. Language in a Bottle: Language Model Guided Concept Bottlenecks for Interpretable Image Classification. *arXiv preprint arXiv:2211.11158*, 2023.

[2] Carlos Busso, Murtaza Bulut, Chi-Chun Lee, Abe Kazemzadeh, Emily Mower, Samuel Kim, Jeannette N. Chang, Sungbok Lee, and Shrikanth Narayanan. IEMOCAP: Interactive Emotional Dyadic Motion Capture Database. *Language Resources and Evaluation*, 42(4):335–359, 2008.

[3] Benjamin Elizalde, Soham Deshmukh, Mahmoud Al Ismail, and Huaming Wang. CLAP: Learning Audio Concepts From Natural Language Supervision. In *Proceedings of Interspeech*, 2023.

[4] Cynthia Rudin. Stop Explaining Black Box Machine Learning Models for High Stakes Decisions and Use Interpretable Models Instead. *Nature Machine Intelligence*, 1(5):206–215, 2019.

[5] Pang Wei Koh, Thao Nguyen, Yew Siang Tang, Stephen Mussmann, Emma Pierson, Been Kim, and Percy Liang. Concept Bottleneck Models. In *Proceedings of the 37th International Conference on Machine Learning*, 2020.

[6] Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning Transferable Visual Models from Natural Language Supervision. In *Proceedings of the 38th International Conference on Machine Learning*, 2021.

[7] Mert Yuksekgonul, Maggie Wang, and James Zou. Post-hoc Concept Bottleneck Models. In *ICLR 2022 Workshop on PAIR2Struct*, 2022.

[8] Tian Yun, Usha Bhalla, Ellie Pavlick, and Chen Sun. Do Vision-Language Pretrained Models Learn Primitive Concepts? *arXiv preprint arXiv:2203.17271*, 2022.

[9] Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J. Liu. Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer. *Journal of Machine Learning Research*, 21(140):1–67, 2020.

[10] Aman Madaan, Niket Tandon, Prakhar Gupta, Skyler Hallinan, Luyu Gao, Sarah Wiegreffe, Uri Alon, Nouha Dziri, Shrimai Prabhumoye, Yiming Yang, Sean Welleck, Bodhisattwa Prasad Majumder, Shashank Gupta, Amir Yazdanbakhsh, and Peter Clark. Self-Refine: Iterative Refinement with Self-Feedback. *Advances in Neural Information Processing Systems*, 36, 2023.

[11] Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric P. Xing, Hao Zhang, Joseph E. Gonzalez, and Ion Stoica. Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. *Advances in Neural Information Processing Systems*, 36, 2023.

[12] George L. Nemhauser, Laurence A. Wolsey, and Marshall L. Fisher. An Analysis of Approximations for Maximizing Submodular Set Functions—I. *Mathematical Programming*, 14(1):265–294, 1978.